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The Role of Social Feedback in Technology Acceptance: A One-Week Diary Study with Exoskeleton Users at the Workplace

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ABSTRACT

Occupational exoskeletons constitute a new assistive technology for industrial workers. In the social context of the workplace, colleagues are likely to react to the noticeable appearance of these devices. Drawing on established technology acceptance theories and recent exoskeleton research, we addressed the following issues: (1) What is the effect of co-worker feedback and self-perceived attractiveness on workers' intention to use an exoskeleton over time? And (2) do these variables predict intention to use beyond perceived usefulness and ease of use? We examined these questions in a one-week diary study with 22 industrial workers. They tested passive exoskeletons during their regular work for six consecutive days and completed a short questionnaire including evaluations of perceived usefulness, ease of use, attractiveness, and social feedback received on a daily basis. The collected data were analyzed using hierarchical linear models (HLM). Results showed that usefulness (but not ease of use) significantly predicted intention to use an exoskeleton. Over and above, self-perceived attractiveness significantly affected initial intention to use, whereas the effects of social feedback unfolded over time. Our study contributes to a better understanding of the social dynamics experienced by users of wearable technology.

KEYWORDS

Exoskeleton; social feedback; attractiveness; technology acceptance; diary field study; workplace

1. Introduction

Social evaluations, such as a critical comment or a “thumbs up” from someone important, can have a strong impact on our mood and self-esteem (Burrow & Rainone, 2017; Vanderhasselt et al., 2015). Considering that many adults spend more day time with colleagues at work than with their family and friends (Kirby et al., 2013), positive social feedback by peers in the workplace plays a prominent role for the motivation and well-being of employees (Bauer et al., 2007). Numerous factors can provoke social reactions at work, including an employee's bodily appearance and the technical gadgets they use. Unusual outfits may elicit appraising or disparaging glances. Carrying a new smartwatch may be seen as “cool” or “uncool” among colleagues (Sundar et al., 2014). As occupational exoskeletons are worn almost like a piece of clothing, they bring the two factors of bodily appearance and technical device together in a novel way.

Current occupational exoskeletons are wearable systems designed to reduce the physical workload of workers, for instance through supporting them in heavy lifting, overhead tasks, or adopting postures that place a lot of strain on the body (Lowe et al., 2019). Their design reflects the human anatomy (Gopura et al., 2011) and may include levers, springs, actuators, sensors, and various types of computing technology (Ruiz-Olaya et al., 2018). Industrial exoskeletons typically feature stiff

components like metal rods and plastic shells, along with elastic textiles and are visible on the worker's body from a distance. Despite empirical research demonstrating health benefits of occupational exoskeletons (Bär et al., 2021), skepticism among workers persists (Bewick & Gardner, 2000). Recent studies have shown that workers are reluctant to use exoskeletons if they feel slowed down by rigid parts on their bodies, do not experience immediate physical relief, or fear extra effort from handling the exoskeleton (Elprama et al., 2023; Siedl & Mara, 2021, 2022).

As with all technologies, the potential benefits of exoskeletons (as promised by manufacturers) can only be realized if workers are willing to use them. The unique ways in which exoskeletons alter the body image necessitate careful examination of user experience and acceptance factors. However, although industrial companies are increasingly offering exoskeletons to their employees (Nussbaum et al., 2019), research into what drives workers' *intention to use* exoskeletons initially and in the long term is still scarce (Siedl & Mara, 2022). For example, how does it affect workers when colleagues comment on their exoskeletons? Does their motivation to wear the exoskeleton decline if they are called “Robocop” or “cyborg” (as exoskeleton testers in industry have reported to us anecdotally)? What if this happens not just once, but on every work day?

To date, there have been very few quantitative studies in which actual workers tested exoskeletons over extended periods and provided feedback on their real-life experiences.

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The present work thus constitutes, to the best of our knowledge, the first diary study aimed at understanding how workers' acceptance of exoskeletons evolves over a full work week. Our analysis goes beyond traditional utilitarian predictors of technology acceptance—namely, usefulness and ease of use. Additionally, we explore previously overlooked social factors, such as the type of feedback workers receive from colleagues and their perceptions of attractiveness while wearing an exoskeleton.

2. Exoskeleton acceptance: Related work

2.1. Pragmatic qualities of exoskeletons

Technology acceptance refers to the demonstrable willingness of a person to use a particular technology for its intended purpose (Dillon & Morris, 1996). According to Ajzen's classic Theory of Planned Behavior (Ajzen, 1991), which remains an important foundation of motivation research, behavior is preceded by corresponding intentions. In the technology acceptance literature, a person's declared *intention to use* a technology is thus considered a particularly relevant prerequisite for its actual use. One widely recognized research model in this field is the Technology Acceptance Model (TAM) by Davis (Davis, 1989). Originally developed in the 1980s for research on IT systems in offices, the model has since been extended and adapted many times for new technological contexts. As a result, a number of alternatives have emerged, including TAM2 (Venkatesh & Davis, 2000), TAM3 (Venkatesh & Bala, 2008), the Unified Theory of Acceptance and Use of Technology (UTAUT by Venkatesh et al., 2003; UTAUT2 by Venkatesh et al., 2012), and highly specialized theoretical models, for instance, for social robots (Heerink et al., 2010) and autonomous vehicles (Hewitt et al., 2019). What most of them have in common is that features related to effort and outcome—also called pragmatic qualities (Hassenzahl, 2018; van Schaik & Ling, 2011)—are considered important predictors of intention to use a technology (Hornbæk & Hertzum, 2017). While the specific terminology might differ across models, the following two variables are commonly seen as key determinants: (1) *Usefulness*, typically operationalized as how much users perceive or expect a system to improve their work performance, and (2) *ease of use* (also referred to as usability), which addresses the perceived or expected effort associated with using the system. Numerous empirical studies support the relevance of performance- and effort-related variables for predicting technology acceptance (i.e., a person's intention to use a system), including meta-analyses in the context of e-health technologies (Calegari et al., 2024) and wearable health trackers (Gopinath et al., 2022).

When shifting focus to occupational exoskeletons, it should thus matter how easy the handling of an exoskeleton appears to users (including donning, doffing, and adjusting) and how much personal productivity gain they expect. Previous empirical work underscores the importance of such pragmatic attributes to exoskeleton acceptance. Applying an adapted version of the UTAUT model, Elprama et al. (2020) were among the first to systematically investigate pre-trial

attitudes of 124 industrial workers to a passive back-assistive device. They found large effects of effort expectancy (i.e., ease of use) and performance expectancy (i.e., usefulness) on intention to use the exoskeleton, with the former having a greater impact than the latter. Several field studies have provided additional evidence of pragmatic attributes impacting workers' willingness to adopt exoskeletons. For instance, Hensel and Keil (2019) provided 30 automotive workers with passive back-supporting exoskeletons for one to four weeks, finding that comfort in work execution, wearing comfort, and usability correlated with higher intentions to use the device. Similarly, Siedl et al. (2021) observed that task-technology fit, wearer comfort, and physical strain relief significantly influenced exoskeleton acceptance among 58 food retail workers testing exoskeletons for up to seven hours during regular work. Schwerha et al. (2022) conducted a qualitative decision-tree analysis based on data from 15 participants using a passive exoskeleton in manufacturing, linking comfort, task-technology fit, and usefulness to exoskeleton acceptance. Additionally, a recent review article by Elprama et al. (2022) emphasized that – despite an overall still limited body of exoskeleton acceptance literature – perceived usefulness and ease of use can be considered empirically meaningful determinants. Against this background, we propose the following first hypothesis:

H1: Usefulness and ease of use predict workers' intention to use an exoskeleton (Model 1).

2.2. Exoskeletons in a social context

While empirical exoskeleton studies have hitherto focused on pragmatic qualities, the wider technology acceptance research community tends to consider a broader range of acceptance factors (Doulani, 2019). Specifically, the social context has been acknowledged as crucial to understanding users' intentions to adopt or reject a technology in work settings (Malhotra & Galletta, 1999; Venkatesh & Morris, 2000). Starting from two fundamental psychological theories—the Theory of Reasoned Action (Fishbein & Ajzen, 1975) and the Theory of Planned Behavior (Ajzen, 1991)—derivates of the TAM incorporated subjective norm (i.e., whether a user thinks that important others approve of a certain action), image (i.e., whether a person believes that technology use is associated with prestige and status), and social influence as variables (Malhotra & Galletta, 1999; Venkatesh & Bala, 2008; Venkatesh & Davis, 2000; Venkatesh & Morris, 2000). Although the TAM and its variants provide important foundations for the study of social acceptance (Koelle et al., 2017; Uhde & Hassenzahl, 2021), related research typically measures attitudes prior to use rather than actual usage experiences (Uhde & Hassenzahl, 2021). Further, many theoretical approaches overlook the heterogeneity of social reactions in a user's environment (Koelle et al., 2020), for example, when some colleagues endorse a technology and others do not. In a workplace that gathers a variety of individuals with unique experiences and needs, the intricate social dynamics at play can lead to

highly diverse evaluations of technology use (Elprama et al., 2022).

In human-computer interaction research, social acceptance is often evaluated by the absence of negative judgment from others (Kelly & Gilbert, 2016). Yet, this concept is multifaceted; users not only interpret how others perceive them but also strive for consistency with their self-image, integrating the social feedback they receive with their self-perceptions (Koelle et al., 2015; 2020; Montero et al., 2010; Sanitioso & Wlodarski, 2004; Williamson, 2011). As a result, users may adjust their technology use or discontinue it altogether, driven by a desire to manage the impression they make on others to maintain a positive self-image (Goffman, 1959).

Although quantitative research on exoskeleton acceptance has largely ignored the role of social impressions, some qualitative studies provide hints that (imagined) judgments from others and concerns about attractiveness may indeed affect a person's willingness to use an exoskeleton (Baltrusch et al., 2020; Cha et al., 2020; Siedl & Mara, 2022). For instance, participants interviewed by Siedl and Mara (2022) expressed worries about appearing “funny”, “weird”, or “less human” while wearing an exoskeleton, whereas others felt that it could make them “look cool”. This shows that – in the context of occupational exoskeletons – there are in fact two facets of users' perceived attractiveness to be considered: First, the concept of physical attractiveness is concerned with the appeal of an individual's bodily appearance to others. This includes attributes such as body proportions (Little et al., 2011; Swami, 2008), which are assumed to impact self-image and other important experiences, such as positive social interaction (S. L. Anderson et al., 2008). Second, attractiveness in research into human-computer interaction, in contrast, usually refers to how aesthetically pleasing a system seems to the user. This perspective integrates both aesthetic and utility considerations (Levy, 2006). Levy (2006) also suggested that the assessment of aesthetics is inherently subjective, but can be grounded in both the individual and in socialized judgments. The notion of attractiveness, as it pertains to exoskeletons, thus extends beyond mere physical appearance and incorporates how the system is perceived by the wearer and others. It encompasses socially accepted styles and preferences, which are essential in judgments of “coolness”, as indicated by research from Sundar et al. (2014) and also with respect to interactive wearables explored by J. Kim and Park (2019). In light of the literature presented, we therefore propose:

H2: Social reactions and perceived personal attractiveness predict workers' intention to use an exoskeleton (Model 2).

H3: Social reactions and perceived personal attractiveness predict workers' intention to use an exoskeleton beyond usefulness and ease of use (Model 3).

2.3. Temporal variations in exoskeleton acceptance

Supposing that a worker's willingness to use an exoskeleton is a binary yes-or-no decision made on a given day without changing thereafter would be a simplistic view. Instead,

acceptance evolves through an ongoing decision-making process, where users repeatedly evaluate their experiences with the technology in their work context (Koelle et al., 2020; Schade & Schlag, 2003; Venkatesh & Bala, 2008), as shown in the fields such as human-robot collaboration (Leichtmann et al., 2024). Factors that initially motivate employees to try a new workplace tool may not necessarily sustain their long-term commitment (Kumar Kakar, 2017).

On the one side, users may be curious and excited about exoskeletons, with a strong intention to use them in initial stages. This may be due to a positive novelty effect (Amandels et al., 2019) or expectations of increased strength and immediate cessation of physical complaints as conveyed by science fiction or media reports (Siedl & Mara, 2021, 2022). Over time, however, sustained use may give rise to a more nuanced perspective and reveal, for instance, limitations in usability and wearer discomfort (Hensel & Keil, 2019), or insufficient benefits for musculoskeletal symptoms (S. Kim et al., 2021).

On the other side, if exoskeletons trigger fears at the beginning (e.g., of not being able to use them properly or of the company expecting higher work performance) or if users are afraid of looking “funny” when wearing an exoskeleton, their initial intention to use may be limited (Siedl & Mara, 2022). However, as users gain experience, they typically grow accustomed to a system (Koelle et al., 2020). Over time, their initial unfamiliarity may transform into a routine assimilation of the technology into their daily work, thereby establishing new habits and gradually increasing acceptance (Stepanovic et al., 2019).

Temporal variations in workers' intention to use exoskeletons may also deviate from linear patterns, as, for example, suggested by the learning-curve effect (Aldunate & Nussbaum, 2013; Glass, 1999): Intention to use may decline when initial enthusiasm is displaced by errors and uncertainties that typically occur during the implementation phase. This could worsen the situation temporarily for workers before improving and leading to higher acceptance levels.

Prior exoskeleton field research has largely ignored potential dynamic shifts in acceptance, not least due to the substantial effort that conducting longitudinal studies typically requires. Many exoskeleton studies even investigated acceptance without any actual human-system interaction or after only a short trial period. Both the importance of studies conducted in authentic working environments of users and the importance of considering the effects of exoskeleton use over time have been highlighted repeatedly in the exoskeleton literature (Howard et al., 2020; Nussbaum et al., 2019). Given the many potential effects of time on exoskeleton acceptance, in addition to hypotheses 1–3, we posit the following research question:

[RQ1] How does workers' intention to use an exoskeleton change over time?

3. Method

To test Hypotheses 1–3 and investigate our Research Question 1, we conducted a diary study with actual industrial workers

who used occupational exoskeletons on six consecutive workdays (excluding weekends). On each test day, participants were required to wear a back- or arm-supporting exoskeleton for at least two hours during their regular working time and to provide feedback by completing a short questionnaire afterwards. Daily worker responses included evaluations of *intention to use*, *perceived usefulness*, *perceived ease of use*, *social reactions*, and *self-perceived attractiveness*. Our research complies with the tenets of the Declaration of Helsinki and was approved by the work council committees of all companies participating. Sociodemographic data was obtained before the exoskeleton trials prior to the study.

Based on learnings from our own previous work and recent exoskeleton studies that found no significant acceptance differences for test durations ranging between 0.5 and 7 hours (Siedl et al., 2021), we assumed daily trials of two hours to be sufficient for novice exoskeleton users to form experience-based impressions. We consulted with occupational health and safety practitioners, work council members, and other company experts who agreed on a two-hour daily test interval as it was found to adequately account for task variability in workplaces on the one hand and considered participants' lack of prior experience with wearing industrial exoskeletons at work on the other hand. A data collection period of six days was the best possible compromise between our scientific requirements and the resource constraints of the companies participating.

3.1. Participants and workplace selection

The field study was conducted with initially $N = 24$ workers in two Austrian companies, one operating in the field of mattress manufacture and one in the distribution of electrical appliances and electrical installation material. By reviewing economic workplace data and in coordination with workplace supervisors and work council members, we identified several workplaces as suitable for trials using either the PAEXO Back or the PAEXO Shoulder exoskeleton. All workplaces selected were found to involve significant strain on the workers' bodies (with a focus on back and arms) resulting from repetitive symmetric and asymmetric lifting and/or the manipulation of loads while extending the arms. Size and weight of the components handled varied to a great extent across (and partly even within) workplaces.

Data from two participants had to be excluded from statistical analyses, as they terminated participation after one day (one person due to a change in work shift and another person due to sick leave). The remaining $n = 22$ workers completed the daily questionnaire on at least two days after using an exoskeleton and thus formed the final sample. They were aged 25 to 56 ($M_{\text{age}} = 39.62$ years, $SD_{\text{age}} = 10.09$) and classified themselves as female or male (45.45% women, 54.55% men, 0.00% non-binary). While 20 participants did not indicate any prior experience with exoskeletons, two workers reported to have tried the technology once.

The final sample consisted of workers who executed different kinds of manual material-handling activities. At the

mattress manufacturing company, three participants were responsible for cutting thick blocks of mattress foam horizontally into layers by means of a circular mattress foam-cutting machine. To this end, the workers lifted the mattress block manually or with a small forklift onto the cutting machine and manually pulled the cut block down onto a pallet truck (WP #1). At another workstation, single layers of mattress foam were put onto a contour-cutting machine. Three participants received finished contour parts after this processing step and manually stacked them next to their workplace (WP #2). To complete the assembly of mattresses, layers of foam and various types of insert materials must be glued together. Using special glue guns, five of the participants carried out gluing activities at the waist level, with their bodies slightly bent forward and hands outstretched. (WP #3). One participant covered finished mattresses and mattress parts with a protective plastic cover on a conveyor belt, mostly in a forward-bent posture with the hands outstretched (WP #4).

At the distributor of electrical appliances, one participant received goods from suppliers, which involved manual lifting and transferring weighty cable-related products from cargo carriers to defined places (WP #5). Six participants were responsible for preparing goods for further transport. Two of these organized goods of up to 5 kg into roller containers, working between ground level and overhead height (WP #6). The other four workers relocated heavy cable material or especially long and bulky goods from storage pallets to defined boxes or storage racks. Despite the use of lifting aids for some tasks, work activities at this workplace placed significant strain on the workers' backs with weights reaching up to 40 kg (WP #7). Order-picking jobs were performed by three workers and typically involved picking goods from the bottom, middle, or top (overhead) of the storage shelves (WP #8) (Figures 1–4).

3.2. The exoskeletons

Two commercially available passive exoskeletons were used in the study: PAEXO Shoulder and PAEXO Back, both developed and commercialized by ottobock.

At 1.9 kg (4.2 lb), the PAEXO Shoulder is a lightweight upper-limb exoskeleton that assists in overhead tasks, as it provides a support torque to the user's arms (i.e., brings relief to the shoulder joint when arms are raised). The support torque varies with the arm elevation angle, such that it reaches its maximum when the upper arms are horizontal (i.e., elevation angle of 90 degrees) and is zero when the arms are dropped and in line with the body (Maurice et al., 2020). For this purpose, the upper arm is enclosed by a shell or bracelet that is linked to a passive joint via an arm bar. Hinge joints connected by a support bar to a spring-based mechanism on the back transfer an adjustable portion of the arm weight to the pelvis via a hip belt.

The PAEXO Back exoskeleton weighs 3.9 kg (8.6 lb) and is engineered to transfer the load from the lower back to the thighs, for example, when lifting heavy objects or working in a forward-bent position. The exoskeleton support



Figure 1. Participant who manually pulled the mattress block from the circular mattress foam-cutting machine down onto a pallet while using a back-assistive exoskeleton (WP #1).

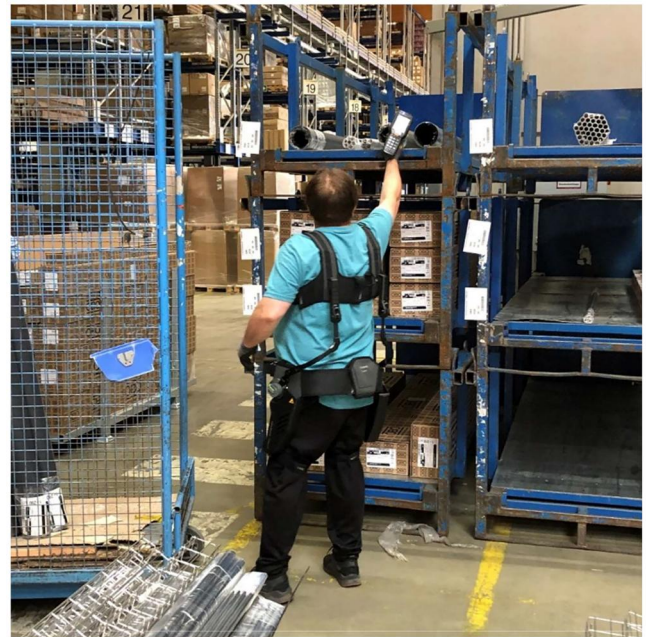


Figure 3. Worker who relocated long and bulky goods from storage pallets to defined boxes or storage racks while wearing a back-assistive exoskeleton (WP #7).



Figure 2. Participant who received finished contour parts from a contour-cutting machine and manually stacked them while wearing a shoulder-supporting exoskeleton (WP #2).

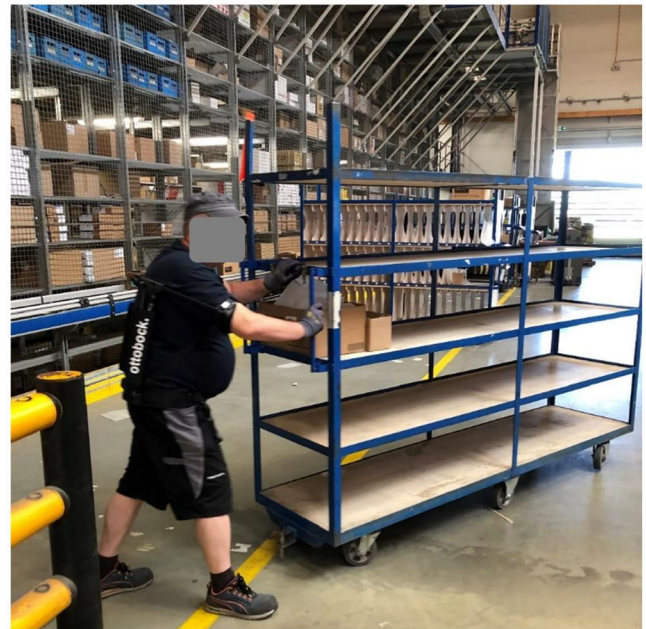


Figure 4. Participant who performed an order-picking job while wearing a back-assistive exoskeleton (WP #8).

structure consists of an upper limb harness connected to a waist belt by means of rigid rods on the back. This adjustable linkage is tied to the waist belt and linked to spring-loaded support elements beside both hips, which are in turn connected to thigh cuffs. When activated by the user, the spring-based support mechanism stores kinetic energy during forward trunk bending and releases it upon straightening up. The hip support moment grows almost linearly with bending (Schmalz et al., 2022). Support is initiated depending on an adjustable angle between the torso and the waist belt. The mechanical control at the hip automatically

switches off when it detects a walking motion. The product website offers further information about the design and functionality of both exoskeletons along with demo videos (<https://ottobockexoskeletons.com/>).

3.3. Measures

We used tablets to provide our participants with the diary questionnaire, including the same measures on all six days. Use of a repeated-measures design required careful

consideration of strategies to mitigate the high drop-out rates typically observed in previous exoskeleton studies (Hensel & Keil, 2019; Siedl & Mara, 2021). Research conducted on web surveys has demonstrated that respondents' willingness to begin and complete a questionnaire decreases with increasing questionnaire length (Galesic & Bosnjak, 2009). Keeping the questionnaire short was therefore assumed to be crucial. A pragmatic solution was to adopt single-item measures, as this ensured minimum completion times. In addition, our questionnaire had to be easily understandable to the workers.

Immediately after each daily test interval, participants were asked to think back to their experience of using the exoskeletons in their workplaces and to assess them using the following measures:

Perceived usefulness and perceived ease of use. Participants rated how useful ("The exoskeleton was useful for my work today.") and easy to use ("The exoskeleton was easy to use today.") they perceived the exoskeleton to be on a 7-point scale (from 1 = *not agree at all* to 7 = *very much agree*).

Social reaction. We assessed daily how other people (e.g., colleagues, supervisors, suppliers) responded to the workers using exoskeletons, asking them to classify the reactions received (e.g., comments, ridicule, amazement) from 1 = *only negative reactions* to 7 = *only positive reactions*.

Perceived personal attractiveness. Participants reported on how much they liked their appearance when wearing the exoskeleton ("I like the way I look with the exoskeleton.") on a 7-point scale (from 1 = *not at all* to 7 = *very much*).

Intention to use. We measured our participants' willingness to continue using the exoskeleton with an adapted item from the validated German Technology Usage Inventory (Kothgassner et al., 2013) ("If I were able to choose, I would continue using the exoskeleton.") on a 7-point scale ranging from 1 = *not agree at all* to 7 = *very much agree*.

3.4. Procedure

In both participating companies, introductory workshops were conducted to facilitate participant recruitment. Within the workshops, we thoroughly explained aim and functionality of the exoskeletons to the workers and informed them comprehensively about the test procedure, which involved completing their regular work tasks while wearing either the PAEXO Shoulder or the PAEXO Back exoskeleton for at least two hours a day over six consecutive days and required completing a brief tablet-based questionnaire after each daily test. We illustrated the risks of the study and highlighted that participation was voluntary. Interested workers enrolled directly with their companies.

Before the exoskeleton trials started, participants gave written informed consent, agreed on data collection, and provided sociodemographic information in a baseline questionnaire (e.g., gender, age, and previous experience with exoskeletons). We showed them how to adjust the exoskeleton to their body size and safely use the device. To verify understanding, the participants practiced proper adjusting

and handling of the exoskeleton in their actual workplaces. However, the study did not include any training period prior to testing the exoskeleton. After two hours working with their exoskeletons, participants completed the daily evaluation questionnaire including four single-item measures (see Measures section) for the first time (t_1) and were asked to repeat the procedure over the next five days (t_{2-6}): putting on the exoskeleton, wearing the device for two hours, returning it, and providing feedback on their experience by using the same evaluation questionnaire. On the last day of the study, five participants had completed the exoskeleton diary on all six days, eight workers had responded on five days, four participants on four days, two on three days, and three on two days. The majority of participants specified organizational circumstances (i.e., time constraints due to work assignments) as the reason for being unable to complete the feedback questionnaire on one or more days. After the trials had been completed, we conducted qualitative interviews with most participants to learn more about the social reactions they had faced as exoskeleton wearers in the workplace. We then fully debriefed them and outlined the subsequent steps following the exoskeleton trials (see Figure 5).

3.5. Data analyses

As the individual worker responses (Level 1; $n = 91$ to 97 data points for each construct) were nested within persons (Level 2; $n = 22$ participants), we applied hierarchical linear models using mixed-method regression analyses in SPSS version 26.0 and a full-information maximum-likelihood algorithm (Bryk & Raudenbush, 1992; Schonfeld & Rindskopf, 2007; Snijders & Bosker, 2012). The null model for the intercept of intention to use revealed significant variance components ($\delta^2_{\text{intercept}} = 1.76$, $p = .006$; $\delta^2_{\text{residual}} = 1.54$, $p < .001$), with an Intraclass-Correlation Coefficient $ICC = .53$. Thus, 53% of the variability in intention to use was due to differences between individual workers, and 47% of the variability can be attributed to fluctuations within an individual worker over the six time points. Gender (0 = male; 1 = female) was a significant Level 2 predictor of intention to use the exoskeleton, $B = -1.93$, $SE = 0.48$, $p < .001$, and accounted for 27.45% of the total variance. Therefore, all statistical models were tested with gender as a control variable.

4. Results

Table 1 summarizes the descriptive statistics and bivariate correlations of the variables studied. Gender – but not age – exhibited relatively high correlations with our variables of interest, with female participants responding more negatively to all constructs. This further justifies the inclusion of gender as a control variable in the analyses. Note that the bivariate correlations were calculated based on participants' mean responses across all time points, without differentiating within- and between-person components, which resulted in relatively high intercorrelations. Levels of significance were set at the standard value of $p < .05$.



Figure 5. Study procedure with two exoskeletons used at workplaces for two hours on six days.

Table 1. Descriptive statistics and spearman correlations of the variables studied ($n = 22$).

Variable	M	SD	Gender	Age	Ease of Use	Usefulness	Social Reactions	Attractiveness
1 Gender								
2 Age	39.62	10.09	-.10					
3 Ease of Use	4.65	1.29	-.01	-.14				
4 Usefulness	3.37	1.46	-.74*	.29	.31			
5 Social Reactions	4.28	1.34	-.63*	-.01	.44*	.68*		
6 Attractiveness	3.72	1.51	-.64*	-.03	.49*	.56*	.66*	
7 Intention to Use	3.14	1.48	-.69*	.12	.40	.79*	.81*	.65*

Note: Gender 0 = male, 1 = female.

* $p < .05$.

To examine RQ1, we tested several alternative models. The first model included the time of measurement (coded in consecutive numbers from day 1 to day 6) as a linear predictor of intention to use the exoskeleton. Results revealed a significant negative effect, $B = -0.16$, $SE = 0.08$, $p = .037$, indicating that, overall, workers' willingness to keep using the exoskeleton decreased significantly over the six days. Furthermore, the significant random variance component of the intercept (see Table 2, Model 0) implies that initial levels in intention to use varied significantly between individuals. To test whether there was not only significant variation in the initial levels of intention to use but also in the trajectories over time, we added the random effect of time to the model. However, the variance component failed to reach significance, $B = 0.14$, $SE = 0.07$, $p = .061$. Thus, the decrease in intention to use over time did not vary significantly between individual workers, but was broadly the same for all. Finally, we also examined whether the trajectory followed a curvilinear trend, but results revealed a nonsignificant quadratic effect of time on intention to use, $B = -0.66$, $SE = 0.36$, $p = .153$. We therefore conclude that intention to use decreased linearly over time and that this decrease was similar across participants (cf. Model 0 in Table 2).

Next, to test our hypotheses 1 to 3, we added the proposed predictors (H1: Ease of use and usefulness, Model 1; H2: Social reactions and attractiveness, Model 2; H3: All, Model 3) to the Basic Model (Model 0). The Basic Model (Model 0) includes the intercept of intention to use as well as its variance component, gender as the control variable, and the linear effect of time (cf. Model 0 in Table 2). We first tested whether the hypothesized predictors had a significant effect on the initial levels of intention to use (Models a; effects on intercept), and then we examined

whether the predictors also had an impact on the trajectories (Models b; effects on slope; i.e., interaction effects between the predictor variable and time). A significant effect of a predictor variable (Models a) indicates that the predictor variable impacts the initial level of intention to use; in contrast, a significant interaction effect between a predictor variable and time (Models b) indicates that the predictor variable determines how intention to use changes over time.

H1 proposes positive effects of ease of use and usefulness on workers' intention to use the exoskeleton. Including ease of use and usefulness to the prediction of intention to use improved the goodness of fit significantly as compared with the basic model (cf. Table 3, comparisons of Models 1 with Model 0). Usefulness, but not ease of use, was significantly associated with higher intention to use in general (cf. Table 2, Model 1a). However, neither of the two predictor variables significantly determined the trajectories of intention to use (cf. Table 2, Model 1b; and Table 3 non-significant increase in model fit of Model 1b as compared with Model 1a).

H2 assumes a positive impact of positive social reactions and self-perceived attractiveness on intention to use. As can be seen in Table 3, these predictor variables also provided a significantly better fit than the basic model (comparisons of Models 2 with Model 0), and Model 2b including the interaction effects with time resulted in a better fit than the simpler Model 2a (comparison of Model 2a with Model 2b). The significant interaction effects (see Table 2, Model 2b) indicate that both social reactions and attractiveness determine the growth trajectories of intention to use the exoskeleton. Furthermore, attractiveness had a significant positive effect on initial levels of intention to use. Figures 6 and 7 illustrate these effects. Workers who experienced more positive social reactions exhibited increasing willingness to

Table 2. Hierarchical linear regression analyses for intention to use the exoskeleton.

Predictors	Model 0		Model 1a		Model 1b		Model 2a		Model 2b		Model 3	
	<i>B</i> (<i>SE</i>)	<i>p</i>	<i>B</i> (<i>SE</i>)	<i>p</i>	<i>B</i> (<i>SE</i>)	<i>p</i>	<i>B</i> (<i>SE</i>)	<i>p</i>	<i>B</i> (<i>SE</i>)	<i>p</i>	<i>B</i> (<i>SE</i>)	<i>p</i>
<i>Basic model</i>												
Intercept	4.49 (0.39)	<.001	1.64 (0.65)	.014	1.47 (0.90)	.107	1.67 (0.96)	.088	1.96 (1.14)	.090	0.46 (1.10)	.679
Gender	−1.91 (0.47)	<.001	−0.65 (0.48)	.188	−0.65 (0.48)	.189	−0.80 (0.51)	.129	−0.96 (0.50)	.064	−0.15 (0.51)	.766
Time	−0.16 (0.08)	.037	−0.07 (0.07)	.315	−0.02 (0.20)	.936	−0.09 (0.08)	.265	−0.12 (0.24)	.623	−0.04 (0.22)	.858
<i>Effects of ease of use and usefulness</i>												
Ease of use			0.03 (0.10)	.730	0.05 (0.18)	.758					−0.03 (0.10)	.778
Usefulness			0.56 (0.10)	<.001	0.58 (0.17)	.001					0.55 (0.10)	<.001
Ease of use x time					−0.01 (0.05)	.893						
Usefulness x time					−0.01 (0.04)	.874						
<i>Effects of social reactions and attractiveness</i>												
Social reactions							0.16 (0.15)	.279	−0.19 (0.21)	.382	−0.28 (0.18)	.139
Attractiveness							0.39 (0.12)	.001	0.74 (0.18)	<.001	0.64 (0.16)	<.001
Social reactions x time									0.14 (0.06)	.034	0.12 (0.06)	.041
Attractiveness x time									−0.15 (0.06)	.010	−0.13 (0.05)	.010
<i>Random variance components</i>												
Intercept	0.86 (0.36)	.017	0.74 (0.33)	.026	0.74 (0.33)	.026	0.72 (0.34)	.033	0.68 (0.31)	.029	0.62 (0.28)	.029
Residuum	1.45 (0.24)	<.001	0.94 (0.16)	<.001	0.95 (0.16)	<.001	1.11 (0.20)	<.001	1.01 (0.18)	<.001	0.73 (0.13)	<.001

Note: Gender 0 = male, 1 = female.

Table 3. Goodness-of-fit statistics and results of nested model comparisons.

Model	Model description	k	Goodness-of-fit			Nested model comparison			
			Deviance	AIC	BIC	Models	χ^2	df	p
M0	Basic model	4	335.805	345.805	358.627				
M1	Ease of use, usefulness								
M1a	on intercept	6	290.148	304.148	321.876	M0-M1a	45.66	2	<.001
M1b	on intercept and slope	8	290.066	308.066	330.860	M0-M1b	45.74	4	<.001
						M1a-M1b	0.08	2	.960
M2	Social reactions, attractiveness								
M2a	on intercept	6	286.037	300.037	317.379	M0-M2a	49.77	2	<.001
M2b	on intercept and slope	8	278.807	296.807	319.103	M0-M2b	57.00	4	<.001
						M2a-M2b	7.23	2	.027
M3	Combined: M1a and M2b	10	247.878	269.878	296.875	M0-M3	87.93	6	<.001
						M1a-M3	42.27	4	<.001
						M2b-M3	30.93	2	<.001

Note: *k* = number of parameters in model. *AIC* = Akaike Information Criterion. *BIC* = Bayesian Information Criterion, with lower values indicating better fit.

further use the exoskeleton over the course of the six days, while participants' intention to use decreased when they experienced more negative social reactions (see Figure 6). 33 responses indicated an excess of positive social feedback, while 17 responses reflected mostly negative feedback from participants' social environment. As examples of negative social feedback they had experienced, some workers reported being called things like "robo-chicken", "bat", and "looking impaired" by co-workers. Other unpleasant experiences included being stared at or being laughed at maliciously. Other exoskeleton testers, in contrast, reported encountering appreciative astonishment ("Wow, cool!") and being called "butterfly" and a "Terminator", which in these cases was experienced as positive feedback. Workers who were named "cyborg" or a "Robocop" when wearing the exoskeleton differed in how they perceived these comments, with some interpreting it negatively and others positively.

Despite the continued importance of social feedback across the six days, the beneficial effect of self-perceived attractiveness when wearing the exoskeleton was present only at the beginning; participants who liked and disliked their appearance wearing the exoskeleton had, respectively, high and low intention to further use the exoskeleton. Over the course of the six days, however, intention to use converged (see Figure 7).

Finally, to examine whether social reactions and attractiveness would add to the prediction of intention to use over and above ease of use and usefulness (H3), we combined Models 1a and 2b into a final Model 3. Model 3 provided a better fit to the data than Models 1a and 2b (cf. Table 3, model comparisons). This indicates that both the "classic" (H1) and the "social" predictors (H2) uniquely contribute to explaining worker intention to use the exoskeleton (see also significant regression coefficients of Model 3 in Table 2). Although the variance components in Model 3 remained significant, a substantial amount of variance could be explained by the predictors. The proportional reduction in prediction error due to all predictor variables in Model 3 is 53% at the within-person level (Level 1), 65% at the between-person level (Level 2), and 59% in total.

5. Discussion

Many jobs require special technical tools – for instance, drills, forklifts, and welding machines – which have become integrated into the working world and help employees to accomplish their tasks. The occupational exoskeleton is a new type of assistive technology for industrial workers, which, due to its high visibility on the wearer's body, is more likely than other tools to elicit reactions from observers

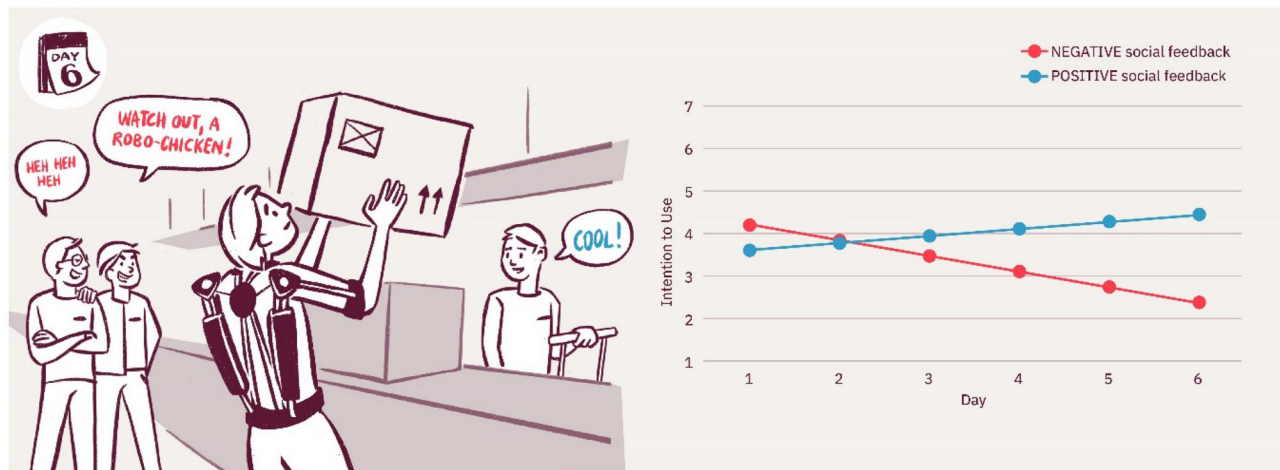


Figure 6. Over time, positive social feedback contributed to higher user acceptance of exoskeletons. The graph on the right illustrates growth trajectories of intention to use for participants experiencing positive versus negative social reactions. *Note.* The slopes were calculated on the basis of the regression coefficients in the final model (Model 3); values for gender were set to 0 (=male), values for ease of use, usefulness, and attractiveness were set to the scale means (value 4), values for positive and negative social reactions were set to 6 (clearly more positive than negative reactions) and 2 (clearly more negative than positive reactions), respectively.



Figure 7. At the beginning, a user's intention to use the exoskeleton was positively influenced by their self-perceived attractiveness with the device. The graph on the right illustrates growth trajectories of intention to use for participants who felt attractive or unattractive wearing the exoskeleton. *Note.* The slopes were calculated on the basis of the regression coefficients in the final model (Model 3); values for gender were set to 1 (=female), values for ease of use, usefulness, and social reactions were set to the scale means (value 4), values for feeling attractive and unattractive were set to 6 and 2, respectively.

and to influence how attractive users feels. However, no previous study has examined whether and how these phenomena affect workers' willingness to wear exoskeletons.

We explored exoskeleton acceptance in a one-week diary study at real industrial workplaces, expanding the limited body of corresponding research. For the first time, we combined perceived pragmatic qualities of exoskeletons as well as attractiveness and social feedback as determinants of user acceptance. Importantly, exoskeleton evaluations in our study did not rely on participants' expectations, but took into account their actual daily experiences while wearing exoskeletons in the social context of the workplace. Focusing on the temporal dimension, our diary-based approach allowed us to better understand dynamic developments in workers' intention to use an exoskeleton over six days, with practical implications for the future design and implementation of occupational wearables.

First, *H1* was partially supported. Perceived usefulness, but not ease of use, significantly predicted intention to use

the exoskeleton in general. If someone perceived the exoskeleton as useful, intention to use was thus generally higher, regardless of whether it was their first day of use or they had been using it for several days. Consistent with earlier studies that identify perceived usefulness as a key determinant of technology acceptance (Elprama et al., 2020; Kim & Park, 2019; Moradbakhti et al., 2023; Venkatesh & Bala, 2008), our results suggest that workers continuously (re-)assess the practical benefits of exoskeletons for their work performance as they gain experience (see also Ma et al., 2023). Contrary to our expectations and some previous empirical findings (Elprama et al., 2020; Venkatesh & Bala, 2008), perceived ease of use did not significantly influence workers' intention to keep using the exoskeleton in our study. This may be due to improvements in the usability of passive exoskeletons over recent years, making aspects such as donning, doffing, and system adjustment relatively straightforward for users (Ziaei et al., 2021). The optimized and well-implemented interface design of commercial

passive exoskeletons might therefore allow drawing attention to new facets of user acceptance. Assuming declining relevance of ease of use, given the increasing simplicity of interfaces, aligns with recent research on fitness apps and artificially intelligent assistants, where ease of use similarly had little impact on user acceptance (Acikgoz et al., 2023; Moradbakhti et al., 2023). Alternatively, it could also be the case that users may not fully understand or consciously consider the intricacies (e.g., customize to fit) of these systems as they use them. In an attempt to cope with the new technology, they deliberately or unconsciously reduce system complexity and assume the technology to automatically conform to user requirements.

Second, *H2* and *H3* were fully supported. Intention to use the exoskeleton was significantly determined by social feedback and self-perceived attractiveness. Interestingly, these two predictor variables contributed differently to intention to use over the course of the six days. Self-perceived attractiveness was particularly important to ensure high acceptance at the very beginning. However, over time, self-perceived attractiveness alone could not keep intention to use the exoskeleton at high levels, and social feedback gained importance for system acceptance instead: Exoskeleton users who experienced positive and negative social reactions exhibited, respectively, higher and lower willingness to further use the exoskeleton.

In the workplace, exoskeleton users are surrounded by many different “actors”, ranging from colleagues and supervisors to suppliers, customers, and third-party players. This creates a social context in which the interaction with exoskeletons takes place (Siedl & Mara, 2022). Social feedback, as an explicit form of verbal and nonverbal articulation of others’ impressions of users, goes beyond “just” what those others might think. It thus addresses an aspect that, to our knowledge, has not been investigated in the context of exoskeleton acceptance. Our results show that participants’ intention to continue exoskeleton use was influenced by how people in their vicinity reacted to them. Responses ranged from comments and inquiries about why users were wearing the systems to nonverbal cues, such as gestures and laughter. Positively experienced feedback led to greater user acceptance over time, while negative feedback led to less acceptance. This may be due to the inherent human need to feel socially connected and supported by others, which is essential for self-motivated behavior and psychological well-being (Ryan & Deci, 2000). In contrast, a lack of social acceptance can put a person at risk for stigmatization (Shinohara & Wobbrock, 2011) and social exclusion, which can negatively affect their sense of self (Bastian & Haslam, 2010). Fostering a supportive social environment is therefore key to overcoming resistance and encouraging continued use of exoskeletons in the workplace. Further, self-perceived attractiveness encouraged greater acceptance of exoskeletons, particularly at the very beginning. Based on the assumption that individuals seek to shape the impression that other persons will form of them (Goffman, 1959), it is unsurprising that how they look is important to them (Baltrusch et al., 2020). Research into technology acceptance has also shown that users’ ideas of what others might think about a

technology can influence whether they accept or reject it, particularly at the very beginning (Venkatesh & Davis, 2000). Additionally, attractiveness can serve as a determinant of social status within groups (C. Anderson et al., 2001). So, if users perceive the exoskeletons to be less aesthetically pleasing, it may lead them to believe that others hold similarly unfavorable opinions of them or the technology itself, potentially influencing their willingness to continue system use. From a practical viewpoint, design choices can thus function as a catalyst for early adoption, allowing the technology to showcase its supportive potential. Involving different target groups in the design process could help to ascertain their needs regarding aesthetics and inform implementation.

Going beyond conformity to societal standards, feeling attractive may also involve a deep appreciation of one’s unique attributes and individuality. Workers’ self-perceived attractiveness, therefore, may be intertwined with enhancement of self-esteem (Kenealy et al., 1991), which reflects a person’s overall evaluation of their self-worth and capabilities (American Psychological Association, n.d.). Exoskeletons might directly impact users’ (physical) self-image because participants, for example, associated themselves with machines and robots while wearing the exoskeletons (Siedl & Mara, 2024), or considered themselves as less likeable (cf. Meyer & Asbrock, 2018). Future research is therefore encouraged to take a closer look at exoskeleton-induced changes in self-image, especially with advanced robotic exoskeletons becoming prevalent for human enhancement.

Third, our results showed that variability in intention to use an exoskeleton was almost equally attributed to (a) differences between users and (b) fluctuations over time. Considering the impact of time on user acceptance, our study highlights the importance of temporal dynamics in predictor variables – as addressed in *RQ1* – and underscores the need for research designs with multiple points of measurement.

Finally, our results also point to the significant role of gender in intention to use an exoskeleton. More than a quarter of the total variance in intention to use can be attributed to user gender. Female participants had significantly lower intention to use and also scored more negatively on all predictor variables. Thus, female workers reported lower usefulness and ease of use, they felt less attractive wearing the exoskeleton, and they received more negative feedback from their co-workers than males. These findings are consistent with multiple empirical studies in which male participants have been found to report higher intention to use various technologies than females (Moradbakhti et al., 2023). One possible explanation for these gender differences could lie in the still prevailing view (even among women) that women are a poorer match for technology than men, which is rooted in classic gender stereotypes portraying females as socially and emotionally oriented and males as action-oriented (Moradbakhti et al., 2022). Related to this are phenomena such as persistent underrepresentation of women in STEM fields (Nimmesgern, 2016) and companies ignoring requirements of potential female users when tailoring their (technical) products specifically to male target

groups (Criado-Perez, 2019). In the same vein, the high proportion of male participants in many previous exoskeleton studies (de Looze et al., 2016; Kermavnar et al., 2021; Moeller et al., 2022) also suggests that the technology may not yet adequately address the needs of women.

Although the present study did not find a significant effect of age on users' intention to use exoskeletons, future research is encouraged to explore the role of age in exoskeleton acceptance more profoundly. For instance, differences in the perceived usefulness of exoskeletons may be attributed to changing physiological conditions with age (Siedl & Mara, 2022). Additionally, Venkatesh et al. (2003) have shown that the perceived effort involved in using a technology moderates its relationship with behavioral intention to use, suggesting that system usability might be particularly important to older user groups. Further, studying social perception of exoskeleton users and identifying potential differences in user groups (younger vs. older workers) could offer new insights into how such technological enhancements intersect with social attitudes and stereotypes (cf. Asbrock, 2010; Meyer & Asbrock, 2018) and inform interventions to promote inclusivity and social acceptance in the workplace.

Having outlined the contributions above, we acknowledge some limitations of our study and suggest possible avenues for future research. First, due to a relatively small sample size, the results must be interpreted with caution. Note, however, that we conducted the study with real exoskeletons in the users' workplaces and achieved a nearly balanced gender distribution. By demonstrating how temporal interaction effects can be taken into account, this study could serve as a starting point for future research in the same vein with a larger number of participants. Second, use of single-item measures was chosen for pragmatic reasons (time constraints faced by the workers, low-literacy target group) in an attempt to find an optimal balance between worker effort and scientific accuracy (Diamantopoulos et al., 2012) in the design of the field study. Third, note that with the concentration on two specific exoskeletons at specific workplaces comes limited generalizability of results. We therefore encourage replicating the study with different exoskeletons in other working environments. Finally, since this study relies solely on individual self-reports, future research into the social acceptance of human-exoskeleton interaction in the workplace could benefit from incorporating both self-assessments and external evaluations, while also exploring the roles of leadership and working climate.

6. Conclusion

Conducting a one-week diary study, we investigated the acceptance of occupational exoskeletons among industrial workers. Our results demonstrate the importance of perceived usefulness, but not ease of use, to workers' intention to continue system use. While a supportive social context was found to be crucial to user acceptance over time, workers' self-perceived attractiveness turned out to be particularly relevant at the very beginning. This suggests that an

aesthetically appealing exoskeleton design might encourage early adoption, while feedback from others can help to foster sustained use. However, gender differences in acceptance raise questions about inclusivity in exoskeleton design. Our insights should inform the design and implementation of exoskeletons to be more responsive to different user groups and inspire future research into human interaction with exoskeletons.

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The authors report there are no competing interests to declare.

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Data availability statement

Data is available online at <https://data.mendeley.com/datasets/f8rfjfnhtz/1>.

References

- Acikgoz, F., Filieri, R., & Yan, M. (2023). Psychological predictors of intention to use fitness apps: The role of subjective knowledge and innovativeness. *International Journal of Human-Computer Interaction*, 39(10), 2142–2154. <https://doi.org/10.1080/10447318.2022.2074668>
- Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2), 179–211. [https://doi.org/10.1016/0749-5978\(91\)90020-T](https://doi.org/10.1016/0749-5978(91)90020-T)
- Aldunate, R., & Nussbaum, M. (2013). Teacher adoption of technology. *Computers in Human Behavior*, 29(3), 519–524. <https://doi.org/10.1016/j.chb.2012.10.017>
- Amandels, S., Eyndt, H. O. h., Daenen, L., & Hermans, V. (2019). Introduction and testing of a passive exoskeleton in an industrial working environment. In S. Bagnara, R. Tartaglia, S. Albolino, T. Alexander, & Y. Fujita (Eds.), *Advances in Intelligent Systems and Computing, Proceedings of the 20th Congress of the International Ergonomics Association (IEA'18)* (820th ed., pp. 387–392). Springer International Publishing. https://doi.org/10.1007/978-3-319-96083-8_51
- American Psychological Association (n.d.). *Self-esteem*. In *APA Dictionary of Psychology*. Retrieved April 1, 2024, from <https://dictionary.apa.org/self-esteem>
- Anderson, S. L., Adams, G., & Plaut, V. C. (2008). The cultural grounding of personal relationship: The importance of attractiveness in everyday life. *Journal of Personality and Social Psychology*, 95(2), 352–368. <https://doi.org/10.1037/0022-3514.95.2.352>

- Anderson, C., John, O. P., Keltner, D., & Krings, A. M. (2001). Who attains social status? Effects of personality and physical attractiveness in social groups. *Journal of Personality and Social Psychology*, 81(1), 116–132. <https://doi.org/10.1037/0022-3514.81.1.116>
- Asbrock, F. (2010). Stereotypes of social groups in Germany in terms of warmth and competence. *Social Psychology*, 41(2), 76–81. <https://doi.org/10.1027/1864-9335/a000011>
- Baltrusch, S. J., Houdijk, H., van Dieën, J. H., & de Kruif, A. J. T. C. M. (2020). Passive trunk exoskeleton acceptability and effects on self-efficacy in employees with low-back pain: A mixed method approach. *Journal of Occupational Rehabilitation*, 31(1), 129–141. <https://doi.org/10.1007/s10926-020-09891-1>
- Baltrusch, S. J., Houdijk, H., van Dieën, J. H., van Bennekom, C. A. M., & de Kruif, A. J. T. C. M. (2020). Perspectives of end users on the potential use of trunk exoskeletons for people with low-back pain: A focus group study. *Human Factors*, 62(3), 365–376. <https://doi.org/10.1177/0018720819885788>
- Bär, M., Steinhilber, B., Rieger, M. A., & Luger, T. (2021). The influence of using exoskeletons during occupational tasks on acute physical stress and strain compared to no exoskeleton – A systematic review and meta-analysis. *Applied Ergonomics*, 94, 103385. <https://doi.org/10.1016/j.apergo.2021.103385>
- Bastian, B., & Haslam, N. (2010). Excluded from humanity: The dehumanizing effects of social ostracism. *Journal of Experimental Social Psychology*, 46(1), 107–113. <https://doi.org/10.1016/j.jesp.2009.06.022>
- Bauer, T. N., Bodner, T., Erdogan, B., Truxillo, D. M., & Tucker, J. S. (2007). Newcomer adjustment during organizational socialization: A meta-analytic review of antecedents, outcomes, and methods. *The Journal of Applied Psychology*, 92(3), 707–721. <https://doi.org/10.1037/0021-9010.92.3.707>
- Bewick, N., & Gardner, D. (2000). Manual handling injuries in health care workers. *International Journal of Occupational Safety and Ergonomics: JOSE*, 6(2), 209–221. <https://doi.org/10.1080/10803548.2000.11076452>
- Bryk, A. S., & Raudenbush, S. W. (1992). *Hierarchical linear models: Applications and data analysis methods*. SAGE Publications.
- Burrow, A. L., & Rainone, N. (2017). How many likes did I get? Purpose moderates links between positive social media feedback and self-esteem. *Journal of Experimental Social Psychology*, 69, 232–236. <https://doi.org/10.1016/j.jesp.2016.09.005>
- Calegari, L. P., R.d. B., & Fettermann, D. C. (2024). A meta-analysis of a comprehensive m-health technology acceptance. *International Journal of Lean Six Sigma*, 15(1), 1–21. <https://doi.org/10.1108/IJLSS-01-2023-0012>
- Cha, J. S., Monfared, S., Stefanidis, D., Nussbaum, M. A., & Yu, D. (2020). Supporting surgical teams: Identifying needs and barriers for exoskeleton implementation in the operating room. *Human Factors*, 62(3), 377–390. <https://doi.org/10.1177/0018720819879271>
- Criado-Perez, C. (2019). *Invisible women: Data bias in a world designed for men*. Abrams Press.
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>
- de Looze, M. P., Bosch, T., Krause, F., Stadler, K. S., & O'Sullivan, L. W. (2016). Exoskeletons for industrial application and their potential effects on physical work load. *Ergonomics*, 59(5), 671–681. <https://doi.org/10.1080/00140139.2015.1081988>
- Diamantopoulos, A., Sarstedt, M., Fuchs, C., Wilczynski, P., & Kaiser, S. (2012). Guidelines for choosing between multi-item and single-item scales for construct measurement: A predictive validity perspective. *Journal of the Academy of Marketing Science*, 40(3), 434–449. <https://doi.org/10.1007/s11747-011-0300-3>
- Dillon, A., & Morris, M. G. (1996). User acceptance of new information technology: Theories and models. *Annual Review of Information Science and Technology*, 14(4), 3–32. <http://hdl.handle.net/10150/105584>
- Doulani, A. (2019). An assessment of effective factors in Technology Acceptance Model: A meta-analysis study. *Journal of Scientometric Research*, 7(3), 153–166. <https://doi.org/10.5530/jscires.7.3.26>
- Elprama, S. A., Bock, S. d., Meeusen, R., Pauw, K. d., Vanderborght, B., & Jacobs, A. (2023). Design and implementation requirements for increased acceptance of occupational exoskeletons in an industrial context: A qualitative study. *International Journal of Human-Computer Interaction*, 1–16. <https://doi.org/10.1080/10447318.2023.2247597>
- Elprama, S. A., Vanderborght, B., & Jacobs, A. (2022). An industrial exoskeleton user acceptance framework based on a literature review of empirical studies. *Applied Ergonomics*, 100, 103615. <https://doi.org/10.1016/j.apergo.2021.103615>
- Elprama, S. A., Vannieuwenhuyze, J. T. A., Bock, S. d., Vanderborght, B., Pauw, K. d., Meeusen, R., & Jacobs, A. (2020). Social processes: What determines industrial workers' intention to use exoskeletons? *Human Factors*, 62(3), 337–350. <https://doi.org/10.1177/0018720819889534>
- Fishbein, M., & Ajzen, I. (1975). Belief, attitude, intention and behavior: An introduction to theory and research. In *Addison-Wesley series in social psychology*. Addison-Wesley.
- Galesic, M., & Bosnjak, M. (2009). Effects of questionnaire length on participation and indicators of response quality in a web survey. *Public Opinion Quarterly*, 73(2), 349–360. <https://doi.org/10.1093/poq/nfp031>
- Glass, R. L. (1999). The realities of software technology payoffs. *Communications of the ACM*, 42(2), 74–79. <https://doi.org/10.1145/293411.293481>
- Goffman, E. (1959). *The presentation of self in everyday life*. Anchor Books.
- Gopinath, K., Selvam, G., & Narayanamurthy, G. (2022). Determinants of the adoption of wearable devices for health and fitness: A meta-analytical study. *Communications of the Association for Information Systems*, 50(1), 445–450. <https://doi.org/10.17705/1CAIS.05019>
- Gopura, R. A. R. C., Kiguchi, K., & Bandara, D. S. V. (2011). A brief review on upper extremity robotic exoskeleton systems. In *International Conference on Industrial and Information Systems (ICIIS'11)* (pp. 346–351). IEEE. <https://doi.org/10.1109/ICIINF.2011.6038092>
- Hassenzahl, M. (2018). The thing and I: Understanding the relationship between user and product. In M. Blythe & A. Monk (Eds.), *Human-Computer Interaction Series. Funology* (Vol. 2, pp. 301–313). Springer International Publishing. https://doi.org/10.1007/978-3-319-68213-6_19
- Heerink, M., Kröse, B., Evers, V., & Wielinga, B. (2010). Assessing acceptance of assistive social agent technology by older adults: The almere model. *International Journal of Social Robotics*, 2(4), 361–375. <https://doi.org/10.1007/s12369-010-0068-5>
- Hensel, R., & Keil, M. (2019). Subjective evaluation of a passive industrial exoskeleton for lower-back support: A field study in the automotive sector. *IIEE Transactions on Occupational Ergonomics and Human Factors*, 7(3–4), 213–221. <https://doi.org/10.1080/24725838.2019.1573770>
- Hewitt, C., Politis, I., Amanatidis, T., & Sarkar, A. (2019). Assessing public perception of self-driving cars [Paper presentation]. Proceedings of the International Conference on Intelligent User Interfaces (IUI '19), In W.-T. Fu, S. Pan, O. Brdiczka, P. Chau, & G. Calvary (Eds.), (pp. 518–527). ACM. <https://doi.org/10.1145/3301275.3302268>
- Hornbæk, K., & Hertzum, M. (2017). Technology acceptance and user experience: A review of the experiential component in HCI. *ACM Transactions on Computer-Human Interaction*, 24(5), 1–30. <https://doi.org/10.1145/3127358>
- Howard, J., Murashov, V. V., Lowe, B. D., & Lu, M. -L. (2020). Industrial exoskeletons: Need for intervention effectiveness research. *American Journal of Industrial Medicine*, 63(3), 201–208. <https://doi.org/10.1002/ajim.23080>
- Kelly, N., & Gilbert, S. (2016). *The WEAR Scale: Developing a measure of the social acceptability of a wearable device* [Paper presentation]. Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI'16), In J. Kaye, A. Druin, C. Lampe, D. Morris, & J. P. Hourcade (Eds.), (pp. 2864–2871). ACM. <https://doi.org/10.1145/2851581.2892331>
- Kenealy, P., Gleeson, K., Frude, N., & Shaw, W. (1991). The importance of the individual in the 'causal' relationship between attractiveness and self-esteem. *Journal of Community & Applied Social Psychology*, 1(1), 45–56. <https://doi.org/10.1002/casp.2450010108>

- Kermavnar, T., Vries, A. W. d., Looze, M. P. d., & O'Sullivan, L. W. (2021). Effects of industrial back-support exoskeletons on body loading and user experience: An updated systematic review. *Ergonomics*, 64(6), 685–711. <https://doi.org/10.1080/00140139.2020.1870162>
- Kim, S., Nussbaum, M. A., Smets, M., & Ranganathan, S. (2021). Effects of an arm-support exoskeleton on perceived work intensity and musculoskeletal discomfort: An 18-month field study in automotive assembly. *American Journal of Industrial Medicine*, 64(11), 905–914. <https://doi.org/10.1002/ajim.23282>
- Kim, J., & Park, E. (2019). Beyond coolness: Predicting the technology adoption of interactive wearable devices. *Journal of Retailing and Consumer Services*, 49, 114–119. <https://doi.org/10.1016/j.jretconser.2019.03.013>
- Kirby, E. L., Wieland, S. M. B., & McBride, M. C. (2013). Work-life conflict. In J. Oetzel & S. Ting-Toomey (Eds.), *The sage handbook of conflict communication* (pp. 377–402). SAGE Publications, Inc. <https://doi.org/10.4135/9781452281988.n16>
- Koelle, M., Ananthanarayan, S., & Boll, S. (2020). *Social acceptability in HCI: A survey of methods, measures, and design strategies* [Paper presentation]. Proceedings of the Conference on Human Factors in Computing Systems (CHI'20), In (pp. 1–19). Association for Computing Machinery. <https://doi.org/10.1145/3313831.3376162>
- Koelle, M., El Ali, A., Cobus, V., Heuten, W., & Boll, S. C. J. (2017). *All about acceptability? Identifying factors for the adoption of data glasses* [Paper presentation]. Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '17), In G. Mark, S. Fussell, C. Lampe, M. Schraefel, J. P. Hourcade, C. Appert, & D. Wigdor (Eds.), (pp. 295–300). ACM. <https://doi.org/10.1145/3025453.3025749>
- Koelle, M., Kranz, M., & Möller, A. (2015). *Don't look at me that way! Understanding user attitudes towards data glasses usage* [Paper presentation]. Proceedings of the International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI'15), In S. Boring, E. Rukzio, H. Gellersen, & K. Hinckley (Eds.), (pp. 362–372). ACM. <https://doi.org/10.1145/2785830.2785842>
- Kothgassner, O. D., Felnhöfer, A., Hauk, N., Kastenhofer, E., Gomm, J., & Kryspin-Exner, I. (2013). *TUI: Technology usage inventory*. ICARUS.
- Kumar Kakar, A. (2017). How do perceived enjoyment and perceived usefulness of a software product interact over time to impact technology acceptance? *Interacting with Computers*, 29(4), 467–480. <https://doi.org/10.1093/iwc/iwx006>
- Leichtmann, B., Meneweger, T., Busch, C., Reiterer, B., Meyer, K., Rammer, D., Haring, R., & Mara, M. (2024). Teaming with a robot in mixed reality: Dynamics of trust, self-efficacy, and mental models affected by information richness. *International Journal of Human-Computer Interaction*, 1–18. <https://doi.org/10.1080/10447318.2024.2331878>
- Levy, S. (2006). *The Perfect Thing: How the iPod shuffles commerce, culture, and coolness*. Simon & Schuster. <https://ebookcentral.proquest.com/lib/kxp/detail.action?docID=5658082>
- Little, A. C., Jones, B. C., & DeBruine, L. M. (2011). Facial attractiveness: Evolutionary based research. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 366(1571), 1638–1659. <https://doi.org/10.1098/rstb.2010.0404>
- Lowe, B. D., Billotte, W. G., & Peterson, D. R. (2019). Astm F48 formation and standards for industrial exoskeletons and exosuits. *IIEE Transactions on Occupational Ergonomics and Human Factors*, 7, 230–236. <https://doi.org/10.1080/24725838.2019.1579769>
- Ma, Z., Gao, Q., & Yang, M. (2023). Adoption of wearable devices by older people: Changes in use behaviors and user experiences. *International Journal of Human-Computer Interaction*, 39(5), 964–987. <https://doi.org/10.1080/10447318.2022.2083573>
- Malhotra, Y., & Galletta, D. F. (1999). *Extending the technology acceptance model to account for social influence: Theoretical bases and empirical validation*. In *Proceedings of the [Paper presentation]*. Annual Hawaii International Conference on Systems Sciences (HICSS'99) (pp. 1006–1019). IEEE. <https://doi.org/10.1109/HICSS.1999.772658>
- Maurice, P., Camernik, J., Gorjan, D., Schirrmeister, B., Bornmann, J., Tagliapietra, L., Latella, C., Pucci, D., Fritzsche, L., Ivaldi, S., & Babic, J. (2020). Objective and subjective effects of a passive exoskeleton on overhead work. *IEEE Transactions on Neural Systems and Rehabilitation Engineering: a Publication of the IEEE Engineering in Medicine and Biology Society*, 28(1), 152–164. <https://doi.org/10.1109/tnsre.2019.2945368>
- Meyer, B., & Asbrock, F. (2018). Disabled or cyborg? How bionics affect stereotypes toward people with physical disabilities. *Frontiers in Psychology*, 9, 2251. <https://doi.org/10.3389/fpsyg.2018.02251>
- Moeller, T., Krell-Roesch, J., Woll, A., & Stein, T. (2022). Effects of upper-limb exoskeletons designed for use in the working environment – A literature review. *Frontiers in Robotics and AI*, 9, 858893. <https://doi.org/10.3389/frobt.2022.858893>
- Montero, C. S., Alexander, J., Marshall, M. T., & Subramanian, S. (2010). *Would you do that?* [Paper presentation]. In Proceedings of the ACM International Conference on Human Computer Interaction with Mobile Devices and Services (MobileCHI'10), M. de Sá, L. Carriço, & N. Correia (Eds.), (pp. 275–278). ACM. <https://doi.org/10.1145/1851600.1851647>
- Moradbakhti, L., Schreibelmayer, S., & Mara, M. (2022). Do men have no need for "feminist" artificial intelligence? Agentic and gendered voice assistants in the light of basic psychological needs. *Frontiers in Psychology*, 13, 855091. <https://doi.org/10.3389/fpsyg.2022.855091>
- Moradbakhti, L., Stiglbauer, B., & Mara, M. (2023). *Predicting technology acceptance: Basic psychological needs as important determinants of intentions to use an AI assistant*. Robopsychology Lab.
- Nimmegern, H. (2016). Why are women underrepresented in STEM fields? *Chemistry (Weinheim an Der Bergstrasse, Germany)*, 22(11), 3529–3530. <https://doi.org/10.1002/chem.201600035>
- Nussbaum, M. A., Lowe, B. D., Looze, M. d., Harris-Adamson, C., & Smets, M. (2019). An introduction to the special issue on occupational exoskeletons. *IIEE Transactions on Occupational Ergonomics and Human Factors*, 7(3–4), 153–162. <https://doi.org/10.1080/24725838.2019.1709695>
- Ruiz-Olaya, A. F., López-Delis, A., & Da Ferreira Rocha, A. (2018). Upper and lower extremity exoskeletons. In J. Segil (Ed.), *Handbook of biomechanics* (pp. 283–317). Elsevier Science & Technology.
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *The American Psychologist*, 55(1), 68–78. <https://doi.org/10.1037/0003-066X.55.1.68>
- Sanitioso, R. B., & Wlodarski, R. (2004). In search of information that confirms a desired self-perception: Motivated processing of social feedback and choice of social interactions. *Personality & Social Psychology Bulletin*, 30(4), 412–422. <https://doi.org/10.1177/0146167203261882>
- Schade, J., & Schlag, B. (2003). Acceptability of urban transport pricing strategies. *Transportation Research Part F: Traffic Psychology and Behaviour*, 6(1), 45–61. [https://doi.org/10.1016/S1369-8478\(02\)00046-3](https://doi.org/10.1016/S1369-8478(02)00046-3)
- Schmalz, T., Colienne, A., Bywater, E., Fritzsche, L., Gärtner, C., Bellmann, M., Reimer, S., & Ernst, M. (2022). A passive back-support exoskeleton for manual materials handling: Reduction of low back loading and metabolic effort during repetitive lifting. *IIEE Transactions on Occupational Ergonomics and Human Factors*, 10(1), 7–20. <https://doi.org/10.1080/24725838.2021.2005720>
- Schonfeld, I. S., & Rindskopf, D. (2007). Hierarchical linear modeling in organizational research. *Organizational Research Methods*, 10(3), 417–429. <https://doi.org/10.1177/1094428107300229>
- Schwerha, D., McNamara, N., Kim, S., & Nussbaum, M. A. (2022). Exploratory field testing of passive exoskeletons in several manufacturing environments: Perceived usability and user acceptance. *IIEE Transactions on Occupational Ergonomics and Human Factors*, 10(2), 71–82. <https://doi.org/10.1080/24725838.2022.2059594>
- Shinohara, K., & Wobbrock, J. O. (2011). *In the shadow of misperception: Assistive technology use and social interactions* [Paper presentation]. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11), In D. Tan, G. Fitzpatrick, C. Gutwin, B. Begole, & W. A. Kellogg (Eds.), (pp. 705–714). ACM. <https://doi.org/10.1145/1978942.1979044>

- Siedl, S. M., & Mara, M. (2021). Exoskeleton acceptance and its relationship to self-efficacy enhancement, perceived usefulness, and physical relief: A field study among logistics workers. *Wearable Technologies*, 2, e10. <https://doi.org/10.1017/wtc.2021.10>
- Siedl, S. M., & Mara, M. (2022). What drives acceptance of occupational exoskeletons? Focus group insights from workers in food retail and corporate logistics. *International Journal of Human-Computer Interaction*, 39(20), 4080–4089. <https://doi.org/10.1080/10447318.2022.2108969>
- Siedl, S. M., & Mara, M. (2024). Am I still human? Wearing an exoskeleton impacts self-perceptions of warmth, competence, attractiveness, and machine-likeness. *Computers in Human Behavior: Artificial Humans*, 100073. <https://doi.org/10.1016/j.chbah.2024.100073>
- Siedl, S. M., Wolf, M., & Mara, M. (2021). *Exoskeletons in the supermarket: Influences of comfort, strain relief and task-technology fit on retail workers' post-trial intention to use* [Paper presentation]. Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction (HRI'21), In C. Bethel, A. Paiva, E. Broadbent, D. Feil-Seifer, & D. Szafir (Eds.), (pp. 397–401). ACM. <https://doi.org/10.1145/3434074.3447200>
- Snijders, T. A. B., & Bosker, R. J. (2012). *Multilevel analysis: An introduction to basic and advanced multilevel modeling* (2nd ed.). Sage.
- Stepanovic, S., Mettler, T., Schmidt-Kraepelin, M., Thiebes, S., & Sunyaev, A. (2019). *Wearable health devices in the workplace: The importance of habits to sustain the use* [Paper presentation]. IEEE Conference on Business Informatics (CBI'19), In (pp. 363–372). IEEE. <https://doi.org/10.1109/CBI.2019.00048>
- Sundar, S. S., Tamul, D. J., & Wu, M. (2014). Capturing “cool”: Measures for assessing coolness of technological products. *International Journal of Human-Computer Studies*, 72(2), 169–180. <https://doi.org/10.1016/j.ijhcs.2013.09.008>
- Swami, V. (2008). The influence of body weight and shape in determining female and male physical appearance. In A. M. Columbus (Ed.), *Advances in Psychology Research* (pp. 195–220). Nova Science Publishers, Inc.
- Uhde, A., & Hassenzähl, M. (2021). *Towards a better understanding of social acceptability*. In [Paper presentation]. Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI'21), Association for Computing Machinery. <https://doi.org/10.1145/3411763.3451649>
- van Schaik, P., & Ling, J. (2011). An integrated model of interaction experience for information retrieval in a web-based encyclopaedia. *Interacting with Computers*, 23(1), 18–32. <https://doi.org/10.1016/j.intcom.2010.07.002>
- Vanderhasselt, M., -Remue, A., Ng, J., Mueller, K. K., Raedt, S. C., & de, R. (2015). The regulation of positive and negative social feedback: A psychophysiological study. *Cognitive, Affective & Behavioral Neuroscience*, 15(3), 553–563. <https://doi.org/10.3758/s13415-015-0345-8>
- Venkatesh, V., & Bala, H. (2008). Technology acceptance model 3 and a research agenda on interventions. *Decision Sciences*, 39(2), 273–315. <https://doi.org/10.1111/j.1540-5915.2008.00192.x>
- Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, 46(2), 186–204. <https://doi.org/10.1287/mnsc.46.2.186.11926>
- Venkatesh, V., & Morris, M. G. (2000). Why don't men ever stop to ask for directions? Gender, social influence, and their role in technology acceptance and usage behavior. *MIS Quarterly*, 24(1), 115–139. <https://doi.org/10.2307/3250981>
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478. <https://doi.org/10.2307/30036540>
- Venkatesh, V., Thong, J. Y. L., & Xu, X. (2012). Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Quarterly*, 36(1), 157–178. <https://doi.org/10.2307/41410412>
- Williamson, J. R. (2011). *Send me bubbles: Multimodal performance and social acceptability* [Paper presentation]. Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI '11), In D. Tan, B. Begole, & W. A. Kellogg (Eds.), (pp. 899–904). ACM. <https://doi.org/10.1145/1979742.1979513>
- Ziaei, M., Choobineh, A., Ghaem, H., & Abdoli-Eramaki, M. (2021). Evaluation of a passive low-back support exoskeleton (Ergo-Vest) for manual waste collection. *Ergonomics*, 64(10), 1255–1270. <https://doi.org/10.1080/00140139.2021.1915502>

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